

**SIMATS ENGINEERING  
ASSESSMENT TOOL 1**

**COURSE NAME: Deep Learning**

**COURSE CODE: MLA0405**

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**COURSE OUTCOME COVERED**

*CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)*

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**QUESTIONS: 5**

**Duration: 1 Hour**

**Total Marks:50**

**Annexure A**

**Analytical Problem Solving**

1. A coin is tossed 10 times and the outcomes are:  
H H T H H H T H H T  
Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).

**ANSWER:**

**Given**

Outcomes:

**H H T H H H T H H T**

Total tosses (**n**) = 10

Number of Heads (**x**) = 7

**Formula**

$$\hat{p} = \frac{x}{n}$$

**Calculation**

$$\hat{p} = \frac{7}{10} = 0.7$$

**Answer**

**Estimated probability of getting Heads = 0.7 (70%)**

2. Out of 50 students, 42 passed an exam.  
Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.

**ANSWER:**

**Given**

Total students = 50

Passed = 42

**Formula**

$$\hat{p} = \frac{x}{n}$$

**Calculation**

$$\hat{p} = \frac{42}{50} = 0.84$$

**Answer**

**MLE of passing probability = 0.84 (84%)**

3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective. Find the MLE for the probability that a bulb is defective.

**ANSWER:**

**Given**

Sample size = 100

Defective bulbs = 8

**Formula**

$$\hat{p} = \frac{x}{n}$$

**Calculation**

$$\hat{p} = \frac{8}{100} = 0.08$$

**Answer**

**Probability that a bulb is defective = 0.08 (8%)**

4. A biased die is rolled 60 times. The number 6 appears 18 times.  
Estimate the probability of rolling a 6 using MLE.

**ANSWER:**

**Given**

Total rolls = 60

Number of sixes = 18

**Formula**

$$\hat{p} = \frac{x}{n}$$

### Calculation

$$\hat{p} = \frac{18}{60} = 0.30$$

### Answer

**Estimated probability of rolling a 6 = 0.30 (30%)**

5. Given: Input (x) = 4

Weight (w) = 2

Actual Output =

10 Prediction: y

=  $w \times x$

1. Find the predicted output.

2. Calculate the error.

3. If the gradient is 4 and learning rate is 0.01, find the updated weight.

### ANSWER:

#### Given

Input (x) = 4

Weight (w) = 2

Actual Output = 10

Learning Rate ( $\eta$ ) = 0.01

Gradient = 4

#### (i) Predicted Output

Formula

$$y = w \times x$$

Calculation

$$y = 2 \times 4 = 8$$

**Predicted Output = 8**

#### (ii) Error

Formula

$$\text{Error} = \text{Actual Output} - \text{Predicted Output}$$

Calculation

$$10 - 8 = 2$$

**Error = 2**

#### (iii) Updated Weight

Gradient Descent Formula

$$w_{\text{new}} = w - \eta \times \text{Gradient}$$

Calculation

$$w_{\text{new}} = 2 - (0.01 \times 4)$$

$$w_{\text{new}} = 2 - 0.04$$

$$w_{\text{new}} = 1.96$$

**Updated Weight = 1.96**

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Code of Conduct:

I **P.AKHILA(192425225)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

P.Akhila

**RUBRICS FOR CALCULATION**

| Criteria                     | Weightage | Level 4 –<br>Exemplary (5<br>Marks)                                                           | Level 3 –<br>Proficient (4<br>Marks)               | Level 2 –<br>Developing<br>(2–3 Marks)              | Level 1 –<br>Beginning<br>(0–1 Mark)                   |
|------------------------------|-----------|-----------------------------------------------------------------------------------------------|----------------------------------------------------|-----------------------------------------------------|--------------------------------------------------------|
| Problem<br>Understanding     | 20%       | Clearly interprets<br>problem, identifies<br>all parameters and<br>requirements<br>accurately | Understands<br>problem with<br>minor gaps          | Partial<br>understanding;<br>misses key<br>elements | Misinterprets<br>or unable to<br>understand<br>problem |
| Method<br>Selection          | 20%       | Selects most<br>appropriate<br>method/formula<br>with strong<br>justification                 | Selects correct<br>method with<br>minor errors     | Inappropriate<br>or partially<br>correct method     | Unable to<br>select method                             |
| Solution<br>Process          | 25%       | Logical,<br>systematic steps<br>with correct<br>approach<br>throughout                        | Mostly correct<br>steps with<br>minor<br>mistakes  | Disorganized<br>steps; multiple<br>errors           | No clear<br>process                                    |
| Accuracy of<br>Calculation   | 20%       | All calculations<br>accurate; correct<br>final answer                                         | Minor<br>calculation<br>errors                     | Frequent errors;<br>incorrect final<br>answer       | No valid<br>calculations                               |
| Interpretation<br>of Results | 15%       | Clearly interprets<br>results with units<br>and engineering<br>relevance                      | Basic<br>interpretation<br>with minor<br>omissions | Limited or<br>unclear<br>interpretation             | No<br>interpretation                                   |